

Assignment 3: HandVis Dataset Analysis

In this study, we hypothesized that because gesture is known to be a type of non-language based communication that can be used as a compensatory channel in the case where the language-based channel is deficient, it should be able to better facilitate communication for users in different modes of second language-involved communication (specifically mixed-proficiency and matching-proficiency in second language). After analyzing the verbal data and idea count results collected from the study, we are able to infer that the varying effects of the different media conditions across language groups demonstrated trends that are both consistent with existing research-supported theory as well as indicative of some conversation-driving effects of the gesture-supporting video interface.

As a precursor statement, for verbal data I have chosen to report results on the analysis of sentence ratio (percentage of sentences spoken by a given user over the course of the dyad's interaction) due to observing that the different measures of speech amount appear to share the same trends across the analyses I happened to perform.

Gender effects across speech amount and idea generation

Overall, regardless of media condition, there was a significant gender effect in sentence ratio (percentage of sentences spoken by given user over the course of the dyad's interaction) for the mixed-proficiency group but not for the matching proficiency group. In EL1-EL2 native speakers, female sentence ratio was ($M = 0.57$, $SE = 0.03$) and male sentence ratio was ($M = 0.63$, $SE = 0.02$) with $p < 0.05$. Second-language speakers in the mixed-proficiency group experienced an even more drastic difference between male and female speech ratio, with female ($M = 0.37$, $SE = 0.02$) and male ($M = 0.51$, $SE = 0.03$), and even smaller $p < 0.001$. This effect can possibly be explained by literature showing that men can be more likely to dominate the conversation, as well as the stronger effect in second language speakers (all from Taiwan) being due to more saliently reinforced gender norms in Eastern versus Western society. Interestingly enough, this effect was not observed in the matching proficiency group, as you can see in Figure 1. Females and males spoke the same amount.

However, this is not to be said for idea counts in the matched-proficiency group. Females in the EL2-EL2 group generated noticeably more ideas when using HandVis than

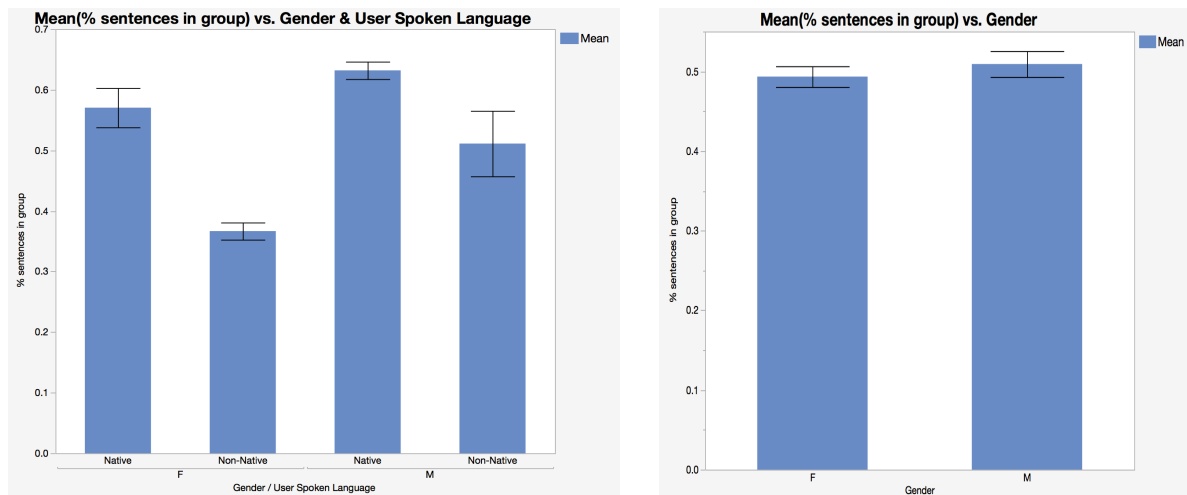


Figure 1. Sentence percentage by gender (and spoken language) for EL1-EL2 and EL2-EL2, respectively

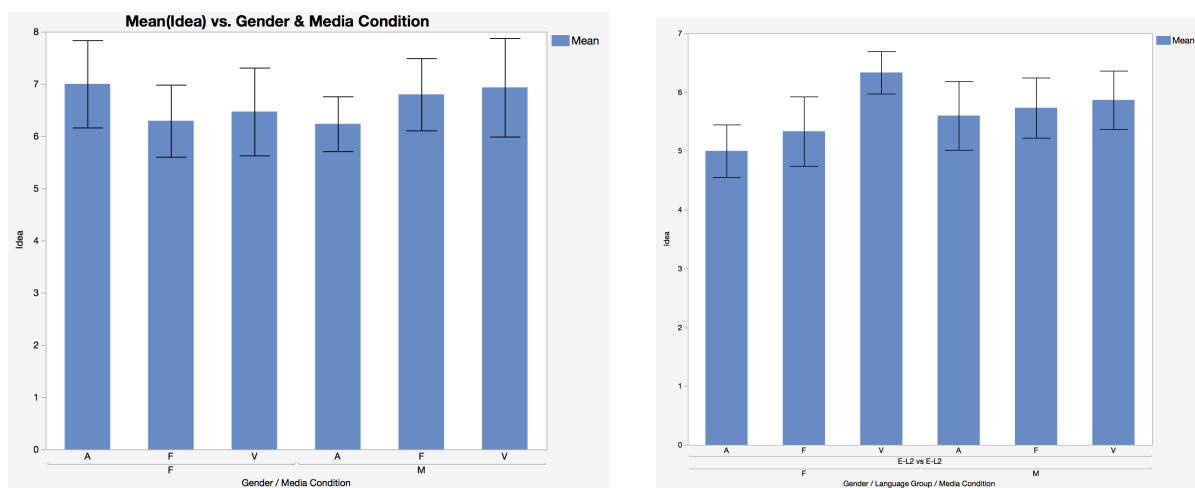


Figure 2. Idea count by gender and media condition for EL1-EL2 and EL2-EL2 respectively

they did in the other two media conditions, while males in EL2-EL2 did not experience any significant effect and nor did either gender in the mixed proficiency group (see Figure 2). Through paired comparison via t-test, the idea generation between HandVis and Audio conditions differed significantly ($p=0.05$) by 1.3 ideas whilst there was an almost-significant difference ($p=0.14$) between HandVis and F2F conditions of 1.0 ideas. Though the reason behind this particular case of improvement by HandVis is unclear, the fact that it is observed only in the EL2-EL2 group may be explained by postulating that the proficiency platform is more equal in EL2-EL2, removing the additional cognitive and social deterrent of suboptimal language proficiency over the existing conversation dominance barrier for the female gender.

On the other hand, a much higher effect of speech amount increase through using HandVis was observed in second-language males in the mixed-proficiency group than for

females. This increase in speech amount through HandVis was not observed overall in the same-proficiency group, and predictable effects demonstrating the media richness theory were observed for native speakers in the mixed-proficiency group (increasing speech amount with richness of media condition). From pairwise t-test for sentence ratio, there was a 18% speech increase from F2F to HandVis ($p=0.2$) and an also large increase from Audio to HandVis of 13% ($p=0.34$). First of all, the larger difference with smaller p-value for the jump from F2F rather than Audio demonstrates that the speech-increasing capabilities of HandVis for second-language speakers may even surpass the channels provided by F2F, which is normally regarded as the gold standard for facilitating the richest channels for communication. Additionally, the fact this is only observable in males and not females may also be due to males not being subject to the existing gender discrepancy observed to be experienced strongly by second-language females in the mixed-proficiency group.

Trends in speech amount aligning with past research and w.r.t. HandVis

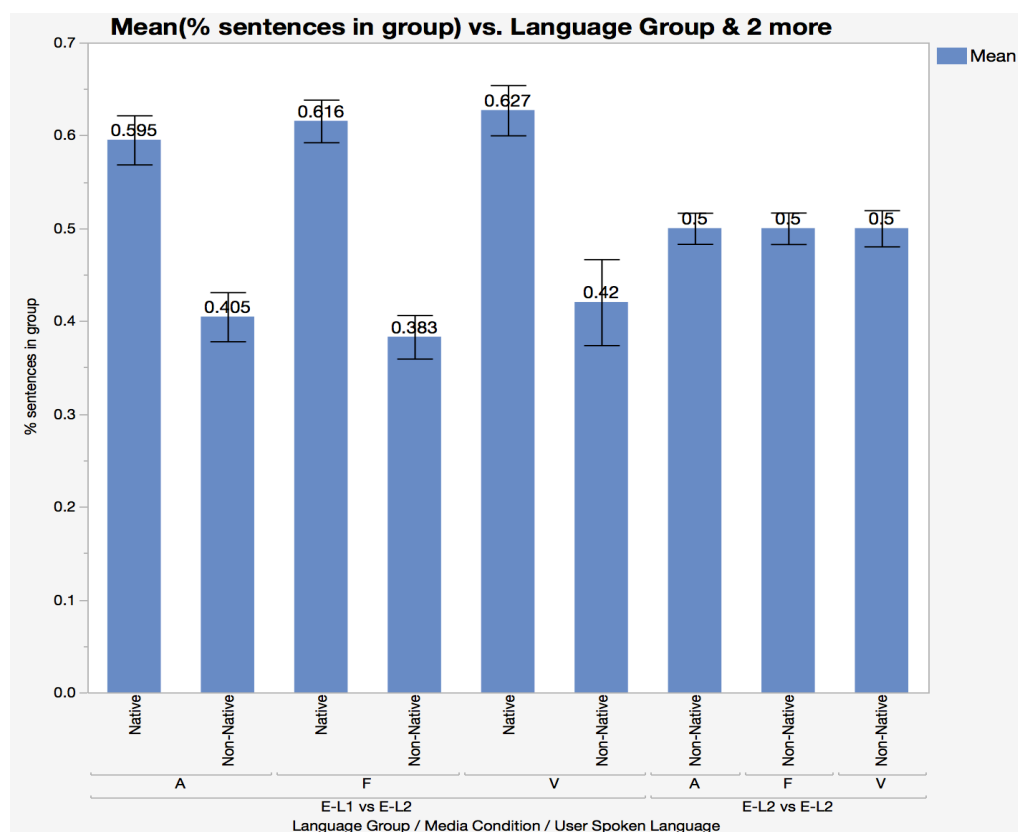


Figure 3. Sentence ratio by Language Group, Media Condition, and Spoken Language

By observing the trends in Figure 3, we can clearly tell that with respect to native and non-speaker interactions the speech production of native speakers, especially in mixed-

proficiency interactions, is significantly higher than that of non-native speakers across all conditions. It is to be expected that non-native speech production in the EL2-EL2 group is higher on average due to more equality on the platform of language proficiency and less issues of discomfort or attribution error. Previous research has also demonstrated that in a mixed-proficiency interaction native speakers overwhelmingly dominate the conversation – this is observed consistently here.

However, though insignificant, the effect of HandVis on increasing speech amount is noticeable for non-native speakers in the mixed-proficiency group. See the table below:

Media Condition	% words in group	% sentences in group
F2F	M = 0.31, SD = 0.14	M = 0.38, SD = 0.1
Audio	M = 0.34, SD = 0.12	M = 0.40, SD = 0.1
HandVis	M = 0.36, SD = 0.19	M = 0.42, SD = 0.19

It is shown here that the speech generation in second-language speakers overall is higher in the HandVis condition than either Audio or F2F, when in the usual case communication is not as well-supported in video conferencing when compared to in-person interaction.

Conclusions

The variance of media condition effects across gender and language proficiency dynamic provide some insight on how such a system as HandVis may interact with various social factors. This can be taken into account for future redesign if we would like to find ways to counter adverse effects from these factors. Additionally, despite the lack of significant and consistent evidence, there is some trend pointing to the possibility of HandVis enhancing both speech production and idea generation capabilities of second-language speakers in both kinds of language groups. The enhancement of these two factors is good news for valuable communication in cross-language contexts since enhancing speech production of the second-language party further levels the playing field in terms of both parties being able to equally express their thoughts to be discussed and synergistically used to form better solutions, and of course increase in idea generation increases the productivity of a brainstorming-centered interaction. These themes seem to be worth further exploration in refinement of future research and system redesign.